

**PREDICTING ACADEMIC SUCCESS AT USM: A COMPUTATIONAL STUDY
USING LINEAR REGRESSION, NAIVE BAYES, AND DECISION TREES**

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ABSTRACT

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This quantitative study developed a multi-model machine learning architecture to predict student academic outcomes using a 10,000-record behavioral dataset. Multiple Linear Regression, Gaussian Naive Bayes, and a Decision Tree classifier evaluated five behavioral predictors to forecast numerical grades and classify academic risk. Algorithmic extraction identified Previous Scores and Hours Studied as the primary predictors of success, while Extracurricular Activities showed minimal statistical impact. Linear Regression achieved the highest accuracy for exact grade forecasting, whereas the Decision Tree excelled in binary risk classification, providing a logical blueprint recommended for integration into the University of Southern Mindanao's academic counseling protocols.

Keywords: Academic Performance, Decision Trees, Educational Data Mining, Linear Regression, Machine Learning, Predictive Analytics.

CHAPTER I

Introduction

The landscape of modern education is undergoing a profound paradigm shift, transitioning from traditional, reactive pedagogical models to proactive, data-driven frameworks. For decades, Higher Education Institutions (HEIs) have relied heavily on summative assessments that evaluate a student's proficiency only after academic terms have concluded. However, the rapid expansion of Educational Data Mining (EDM) and Machine Learning (ML) has introduced the capability to process massive volumes of human behavioral data, fundamentally changing how academic success is understood and managed (Baker & Yacef, 2009; Romero & Ventura, 2010).

By treating scholastic performance not as a random occurrence but as a calculable output derived from quantifiable inputs such as cumulative study duration, sleep cycles, and the completion of practice materials, it is now mathematically feasible to map the precise trajectory of a student's success. Recent systematic reviews in academic prediction emphasize that combining multidimensional behavioral predictors with machine learning models significantly improves operational efficiency and personalized learning interventions in educational decision-making (Daza et al., 2022). Predictive modeling allows institutions to anticipate academic outcomes before formal

examinations take place, transforming raw behavioral data into actionable educational intelligence.

Background of the Study

Within the academic ecosystem of the University of Southern Mindanao (USM), maintaining high academic standards and securing scholarship retention are paramount concerns for the student body. Students are continuously tasked with balancing academic rigor against various behavioral and physiological variables, including sleep deprivation, extracurricular commitments, and independent study hours.

Scientific literature extensively corroborates the impact of these variables. Research demonstrates that behavioral characteristics are strictly interconnected and heavily dictate a student's cognitive functioning; for instance, poor sleep quality actively increases stress and hinders learning capacities, subsequently damaging academic performance (Putra et al., 2023; Ursache et al., 2021). Furthermore, recent machine learning analyses of student study habits reveal that daily study hours exhibit an overwhelmingly strong mathematical correlation ($r = 0.825$) with final exam scores, acting as the dominant predictor of success over demographic factors.

Historically, when students struggle to balance these factors, administrators have limited avenues for early intervention. Addressing this

requires an anticipatory grading model. Utilizing algorithms such as Linear Regression, Naive Bayes, and Decision Trees has proven highly effective for processing educational data to generate accurate forecasts (Harvey & Kumar, 2019). This capstone project leverages a robust dataset of 10,000 distinct student records to train these supervised learning algorithms. By shifting the perspective of academic monitoring from a state of post-exam observation to a state of predictive foresight, this research aims to build a computational architecture capable of identifying the multidimensional correlations between a student's daily habits and their final academic yield.

Statement of the Problem

The primary challenge this research addresses is the inability of current academic monitoring systems to accurately and computationally forecast a student's final Performance Index prior to formal evaluation. While it is generally understood that factors like study habits and sleep influence grades, there is a distinct lack of mathematical quantization regarding how much these variables alter the outcome when evaluated collectively.

Specifically, this computational study seeks to answer the following technical questions:

1. How accurately can a Multiple Linear Regression model predict a student's continuous Performance Index given a multi-variable matrix of

behavioral data (i.e., hours studied, previous academic scores, sample papers practiced, and sleep hours)?

2. To what degree of statistical accuracy can probabilistic and tree-based classifiers (Naive Bayes and Decision Trees) categorize students into binary academic thresholds (Pass/Fail) based on these continuous metrics and categorical inputs (e.g., extracurricular involvement)?
3. Among the deployed machine learning algorithms, which computational model demonstrates the lowest margin of error and the highest predictive reliability when processing a large-scale, 10,000-record dataset?

Objectives of the Study

The central objective of this capstone project is to design, implement, and evaluate a multi-model predictive framework capable of forecasting student academic outcomes. To achieve this, the specific technical objectives are:

- To formulate a continuous predictive model: Implement a Multiple Linear Regression algorithm to calculate exact performance scores based on continuous behavioral features.
- To develop classification frameworks: Train a Naive Bayes classifier utilizing conditional probability, alongside a Decision Tree algorithm utilizing Gini Impurity, to classify students into distinct risk categories.

Previous studies highlight that tree-based models and Naive Bayes often yield high accuracy rates when properly trained on behavioral features (Oyededeji et al., 2020).

- To execute large-scale data processing: Clean, encode, and process a 10,000-row dataset, utilizing a rigorous 80/20 train-test split to ensure the models achieve genuine mathematical generalization rather than algorithmic overfitting.
- To generate comprehensive visual analytics: Produce multidimensional data visualizations—including feature correlation heatmaps, regression scatter planes, and logic-path tree diagrams—to translate complex algorithmic computations into interpretable academic insights.

Significance of the Study

The outputs of this computational research present substantial utility to multiple stakeholders:

- For the CCIT Faculty and University Administration: The resulting models provide the foundational architecture for an automated early-warning system. This allows educators to computationally identify and intervene with at-risk students weeks before final assessments, thereby improving overall institutional retention rates.

- For the Student Body: By establishing the exact statistical weight of variables like Sleep Hours versus Extracurricular Activities, students are provided with mathematically proven blueprints for optimizing their time management.
- For the Field of Computational Science: This research contributes empirical evidence regarding the comparative efficiency of probabilistic versus regression-based machine learning models. As noted by academic researchers, machine learning models frequently offer more accurate predictions and insights into student behavior than standard statistical methods (Maia et al., 2023).

Scope and Delimitation

This study is exclusively focused on the computational prediction of academic performance utilizing supervised machine learning models. The research is strictly bounded by the parameters of the utilized 10,000-record dataset. The independent variables evaluated are mathematically limited to Hours Studied, Previous Scores, Extracurricular Activities, Sleep Hours, and Sample Question Papers Practiced, with the dependent target variable being the continuous Performance Index.

This study explicitly delimits qualitative socioeconomic factors, psychological health metrics, and demographic profiling (such as age, gender,

or financial status), as these variables are not present within the foundational dataset. Furthermore, while the system will output predictive data, the actual deployment of a live software application for university-wide use falls outside the scope of this algorithmic analysis.

CHAPTER II

Algorithm Selection and Computational Rationale

In computational data science, the selection of a machine learning algorithm must be mathematically aligned with the dimensionality and variance of the target dataset. The 10,000-record student performance dataset presents a hybrid computational challenge: it requires both the prediction of a continuous numerical output and the classification of discrete binary outcomes. To address this, a multi-model architecture was selected, deploying algorithms that handle data through fundamentally different mathematical paradigms.

For the prediction of the continuous Performance Index, a deterministic approach is required. Multiple Linear Regression was selected because it efficiently maps the linear correlations between multi-dimensional continuous inputs (such as hours studied and sleep duration) and a single numerical output, providing highly interpretable feature weights.

Conversely, for binary classification (predicting a definitive "Pass" or "Fail"), deterministic regression is computationally inefficient. Therefore, the system deploys two distinct classification paradigms for cross-validation:

1. **A Probabilistic Paradigm:** Gaussian Naive Bayes was selected to process the data through conditional probability, calculating the statistical likelihood of failure based on behavioral habits.
2. **A Non-Parametric Paradigm:** The Decision Tree algorithm was selected to process the data hierarchically. Unlike regression or probability, it maps exact logical thresholds (e.g., minimum required sleep) using impurity metrics, ensuring the model's decision-making process mimics human logical deduction.

By deploying this tri-algorithm approach, the study ensures robust predictive reliability across the dataset, cross-verifying geometric, probabilistic, and logical models.

Mathematical and Theoretical Framework

To transition from theoretical observation to computational execution, the academic problem must be abstracted into distinct mathematical formulas. This study utilizes three distinct algorithmic architectures, each designed to process the five independent behavioral variables: Hours Studied (X_1), Previous Scores (X_2), Extracurricular Activities (X_3), Sleep Hours (X_4), and Sample Question Papers Practiced (X_5).

Multiple Linear Regression (Continuous Prediction)

To forecast the exact numerical value of the Performance Index (Y), the system employs a Multiple Linear Regression (MLR) model. MLR calculates the linear relationship between the continuous target variable and the independent behavioral predictors by assigning an optimized mathematical weight (coefficient) to each feature.

The predictive equation is formalized as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 +$$

Where:

- Y represents the predicted Performance Index.
- β_0 is the y-intercept (the theoretical baseline score if all behaviors were zero).
- $\beta_1, \beta_2, \dots, \beta_5$ represent the calculated regression coefficients, dictating the proportional impact of each respective behavioral feature.
- ϵ represents the stochastic error term, accounting for variance not explained by the model.

To optimize the β coefficients, the algorithm minimizes the Mean Squared Error (MSE) during the training phase across the 10,000 records. The loss function minimized by the computational model is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Where n is the total number of observations, Y_i is the actual recorded score, and $\hat{Y}_i$ is the model's predicted score.

Naive Bayes Classification (Probabilistic Thresholds)

To classify students into binary outcomes (Pass vs. Fail), a Naive Bayes classifier is utilized. This algorithm is rooted in Bayes' Theorem, computing the conditional probability of an academic outcome given a specific set of observed behavioral features. Because the dataset contains continuous variables (like Sleep Hours and Previous Scores), the system specifically employs Gaussian Naive Bayes, which assumes the continuous data follows a normal (Gaussian) distribution.

The foundational Bayes' Theorem is expressed as:

$$P(C_k|x) = \frac{P(C_k) \prod_{i=1}^n P(x_i|C_k)}{P(x)}$$

Where:

- C_k is the target class category (e.g., C_1 = "Pass", C_0 = "Fail").
- x is the vector of observed behavioral features $(X_1, X_2, \dots, X_5)$.

- $P(C_k)$ is the prior probability of the class occurring in the 10,000-record dataset.
- $P(x_i|C_k)$ is the likelihood of observing feature x_i given that the class is C_k .

For the continuous behavioral data, the likelihood is computed using the Gaussian density function:

$$P(x_i|C_k) = \frac{1}{\sqrt{2\pi\sigma_{C_k}^2}} \exp\left(-\frac{(x_i - \mu_{C_k})^2}{2\sigma_{C_k}^2}\right)$$

This formula calculates the probability density by utilizing the mean (μ) and variance (σ^2) of each feature within the training set, allowing the algorithm to mathematically weigh how severe deviations in habits (e.g., extreme sleep deprivation) alter the probability of failure.

Decision Tree Logic (Non-Parametric Node Splitting)

Unlike regression and probabilistic modeling, the Decision Tree algorithm employs a non-parametric, hierarchical structure to partition the student data. The algorithm recursively splits the 10,000 records into smaller, more homogenous subsets based on specific feature thresholds (e.g., splitting students who study > 4 hours from those who study ≤ 4 hours).

To mathematically determine which behavioral feature provides the optimal split at any given node, the model utilizes Gini Impurity. The algorithm

evaluates the purity of potential splits and selects the feature that minimizes impurity, expressed as:

$$Gini(D) = 1 - \sum_{i=1}^c (p_i)^2$$

Where:

- D represents the dataset at the current node.
- c is the total number of classes (2: Pass and Fail).
- p_i is the ratio of students belonging to class i within that specific node.

The model evaluates every possible split across all variables, selecting the split that maximizes Information Gain, formally defined as the reduction in Gini Impurity:

$$Gain(D, A) = Gini(D) - \sum_{v \in Values(A)} \frac{|D_v|}{|D|} Gini(D_v)$$

By executing this calculation iteratively, the computational model constructs a transparent, logical flowchart of the exact behavioral conditions required to pass, satisfying the requirement for algorithmic explainability.

Conceptual and Computational Architecture

To seamlessly process the mathematical models, the system architecture operates on an Input-Process-Output (IPO) framework.

- **Input Stage:** The system ingests the raw, 10,000-row CSV file containing the five behavioral features.
- **Process Stage (Preprocessing & Training):** The categorical variable Extracurricular Activities (Yes/No) is computationally encoded into a binary format (1/0) to satisfy the algebraic requirements of the algorithms. Furthermore, the continuous Performance Index is transformed into a binary threshold (e.g., scores ≥ 60 classified as 1, scores < 60 as 0) to train the classification models. The data is then subjected to an 80/20 train-test split, reserving 2,000 untouched records to validate the mathematical accuracy of the trained weights.
- **Output Stage:** The system generates the finalized Regression coefficients, Gaussian probabilities, Tree logic paths, and evaluation metrics (MSE and Accuracy grids).

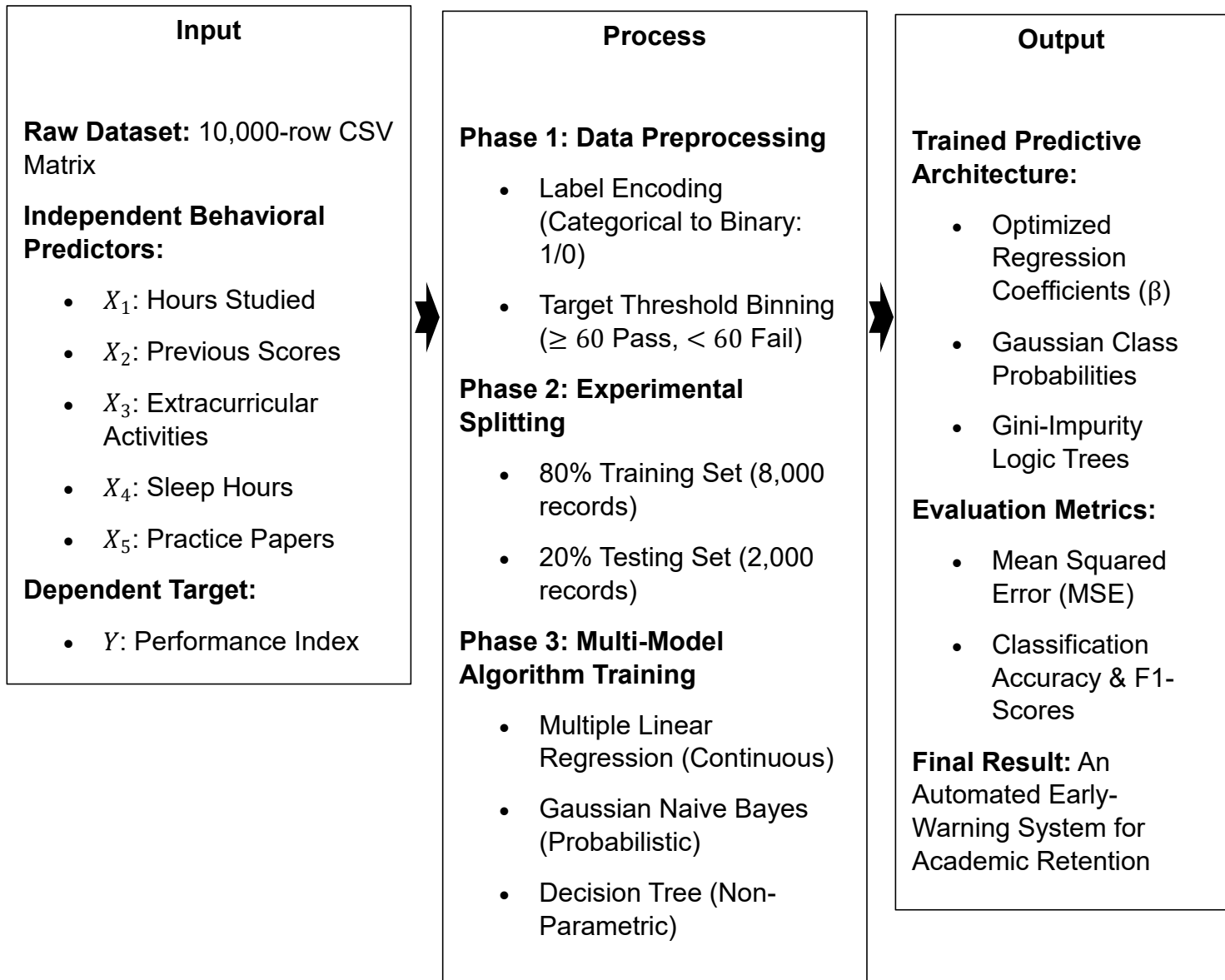


Figure 2.1: Input-Process-Output (IPO) Architecture of the Predictive System

CHAPTER III

Research Design and Computational Environment

This capstone project utilizes a quantitative, computational experimental design to evaluate the predictive efficacy of multiple machine learning models. The research relies on empirical data processing rather than qualitative surveys, focusing entirely on the algorithmic extraction of behavioral patterns from a high-dimensional dataset.

To ensure the models execute with maximum efficiency and minimal latency during the training phases, a robust computational architecture was established. The experiments were conducted on a system equipped with a 13th Generation Intel Core i5 processor, a dedicated NVIDIA RTX 3050 graphics processing unit, and 8GB of active memory. The software environment was built on Python 3.10+, utilizing Visual Studio Code (VS Code) as the primary Integrated Development Environment (IDE). Within this environment, a Jupyter Notebook kernel was deployed to allow for interactive, cell-by-cell data processing and immediate graphical rendering. The core machine learning algorithms and mathematical operations were executed using the scikit-learn (Pedregosa et al., 2011), pandas (McKinney, 2010), pandas, and NumPy libraries, which are industry standards for Educational Data Mining (EDM) implementations.

Table 1: Specifications of the Computational Environment

Component Category	Specification	Primary Function in Experiment
Processor (CPU)	13th Gen Intel Core i5 (13-420H)	Handles core algorithmic calculations and model fitting operations.
Graphics (GPU)	NVIDIA GeForce RTX 3050	Accelerates data processing and visualization rendering.
Active Memory (RAM)	8GB	Manages the 10,000-row dataframe in active memory without bottlenecking.
Programming Language	Python 3.10+	Serves as the foundational syntax for all machine learning operations.
Development Environment	Visual Studio Code (VS Code) with Jupyter Extension	Provides the primary coding interface, enabling interactive cell execution for data

		modeling and visualization.
Core Libraries	Scikit-learn, Pandas, NumPy	Executes the ML algorithms, dataframe manipulation, and matrix math.

Data Gathering Procedure

The primary dataset utilized for this computational experiment was acquired from the Kaggle data repository, specifically selected for its volume and structural integrity. The dataset comprises exactly 10,000 distinct records of student academic behavior, ensuring that the machine learning models have a sufficiently large sample size to discover statistically significant correlations. By utilizing an open-source, pre-validated dataset, the study eliminates the inherent biases and missing values often associated with manual academic surveys, providing a mathematically clean foundation for algorithmic training.

Data Preprocessing and Feature Engineering

Raw data must be structurally optimized before it can be processed by machine learning algorithms, as mathematical models require strict numerical

matrices. The data preprocessing phase involved several critical transformations to convert the human-readable dataset into a machine-readable format.

First, the dataset was loaded into a pandas DataFrame for structural inspection. The categorical variable Extracurricular Activities, initially recorded as string values ("Yes" and "No"), was computationally encoded into a binary numerical format. Using label encoding, "Yes" was mapped to 1 and "No" was mapped to 0, allowing the regression and classification models to weigh this feature algebraically.

Second, to satisfy the distinct requirements of both continuous and categorical algorithms, the target variable was engineered into two separate formats:

- **Continuous Target (For Multiple Linear Regression):** The Performance Index was maintained in its original, continuous numerical state (ranging from 10 to 100) to allow the regression model to predict exact scores.
- **Binary Target (For Naive Bayes and Decision Trees):** A new categorical column, Academic_Status, was derived from the Performance Index. A strict mathematical threshold was established: records with a score of ≥ 60 were classified as 1 ("Pass"), while scores < 60 were classified as 0 ("Fail"). This transformation translates the

continuous grading spectrum into a binary classification problem suitable for probabilistic logic.

Table 2: Dataset Feature Description and Preprocessing Actions

Feature / Column Name	Original Data Type	Preprocessing Action / Transformation	Algorithmic Role
Hours Studied	Integer	Maintained as numerical input.	Independent Predictor (X_1)
Previous Scores	Integer	Maintained as numerical input.	Independent Predictor (X_2)
Extracurricular Activities	String (Yes/No)	Label Encoded: Converted "Yes" to 1 and "No" to 0.	Independent Predictor (X_3)
Sleep Hours	Integer	Maintained as numerical input.	Independent Predictor (X_4)
Practice Papers	Integer	Maintained as numerical input.	Independent Predictor (X_5)

Performance Index	Float (Continuous)	Maintained as numerical input. Binned (≥ 60 Pass, < 60 Fail) for Naive Bayes and Decision Tree.	Target Variable (Y)
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Experimental Setup and Data Splitting

To prevent the algorithms from simply memorizing the dataset (a critical error known as algorithmic overfitting), the 10,000 records were partitioned using a standard 80/20 train-test split. The `train_test_split` function from the scikit-learn library was utilized to randomly divide the dataset into two mutually exclusive matrices.

The feature matrix $X \in \mathbb{R}^{10000 \times 5}$ and the target vector $Y \in \mathbb{R}^{10000}$ were divided as follows:

- **Training Set (80%):** A matrix of 8,000 student records was fed into the algorithms. This subset served as the mathematical foundation for the models to adjust their internal weights, calculate probabilities, and determine Gini Impurities.

- **Testing Set (20%):** A matrix of 2,000 untouched student records was reserved strictly for evaluation. The trained models were required to predict the outcomes of these unseen records, simulating a real-world predictive scenario and providing a mathematically sound metric for accuracy.

A fixed random state (e.g., `random_state=42`) was initialized during the splitting process to ensure deterministic reproducibility, meaning the exact same 8,000 training rows are selected during every execution of the script.

Table 3: Dataset Partitioning Matrix (Train-Test Split)

Data Subset	Percentage Allocation	Number of Records	Function in the Machine Learning Pipeline
Total Dataset	100%	10,000 rows	The complete Kaggle dataset utilized for the study.
Training Set	80%	8,000 rows	Fed into the algorithms to calculate β

			coefficients, Bayesian probabilities, and Gini Impurities.
Testing Set	20%	2,000 rows	Strictly hidden from the models during training; utilized solely to test the accuracy of the predictions.

Algorithm Implementation and Execution

With the data successfully preprocessed and partitioned, the core machine learning models were initialized and executed.

Multiple Linear Regression Execution

The continuous feature matrix was passed into the `LinearRegression()` class. The algorithm iterated over the 8,000 training records, utilizing ordinary least squares optimization to calculate the optimal β coefficients for study hours, sleep, practice papers, extracurriculars, and previous scores. Once trained,

the `predict()` function was called on the 2,000-record testing set to generate an array of exact forecasted performance scores.

Gaussian Naive Bayes Execution

To predict the binary `Academic_Status`, the `GaussianNB()` classifier was instantiated. The algorithm computed the mean and variance for each of the five behavioral features across the passing and failing classes within the 8,000-record training set. Using these calculated parameters, the model applied the Gaussian probability density function to the 2,000 testing records, outputting a discrete array of 1s and 0s based on the highest calculated posterior probability.

Decision Tree Classifier Execution

The `DecisionTreeClassifier()` was implemented to construct the logical flowchart of academic success. To prevent the tree from growing overly complex and memorizing specific noise within the training data, hyperparameter tuning was applied. The `max_depth` parameter was constrained, and the criterion was explicitly set to minimize Gini Impurity. The algorithm evaluated all behavioral thresholds, placing the feature with the highest Information Gain at the root node, and subsequently split the 8,000

records into terminal leaf nodes representing a final "Pass" or "Fail" classification.

CHAPTER IV

Evaluation Metrics

To quantitatively assess the performance of the implemented machine learning models, specific statistical metrics were calculated against the 2,000-record testing set. The models were evaluated based on their respective algorithmic goals: continuous forecasting and binary classification.

For the Multiple Linear Regression model, predictive accuracy was measured using Mean Squared Error (MSE) and the R-squared (R^2) coefficient. The R^2 value indicates the proportion of the variance in the final Performance Index that is predictable from the students' behavioral habits.

For the Gaussian Naive Bayes and Decision Tree classifiers, performance was evaluated using a standard classification matrix, yielding Accuracy, Precision, Recall, and F1-Scores. Accuracy measures the overall percentage of correct Pass/Fail predictions, while the F1-Score provides a harmonic mean of Precision and Recall, ensuring the model's reliability is not artificially inflated by class imbalances within the dataset.

Visualizing the Findings

Data visualization was executed using the matplotlib (Hunter, 2007) and seaborn libraries within the Visual Studio Code (VS Code) environment utilizing a Jupyter kernel. This computational setup was required to translate the high-dimensional mathematical outputs into interpretable, human-readable formats.

Feature Correlation Heatmap

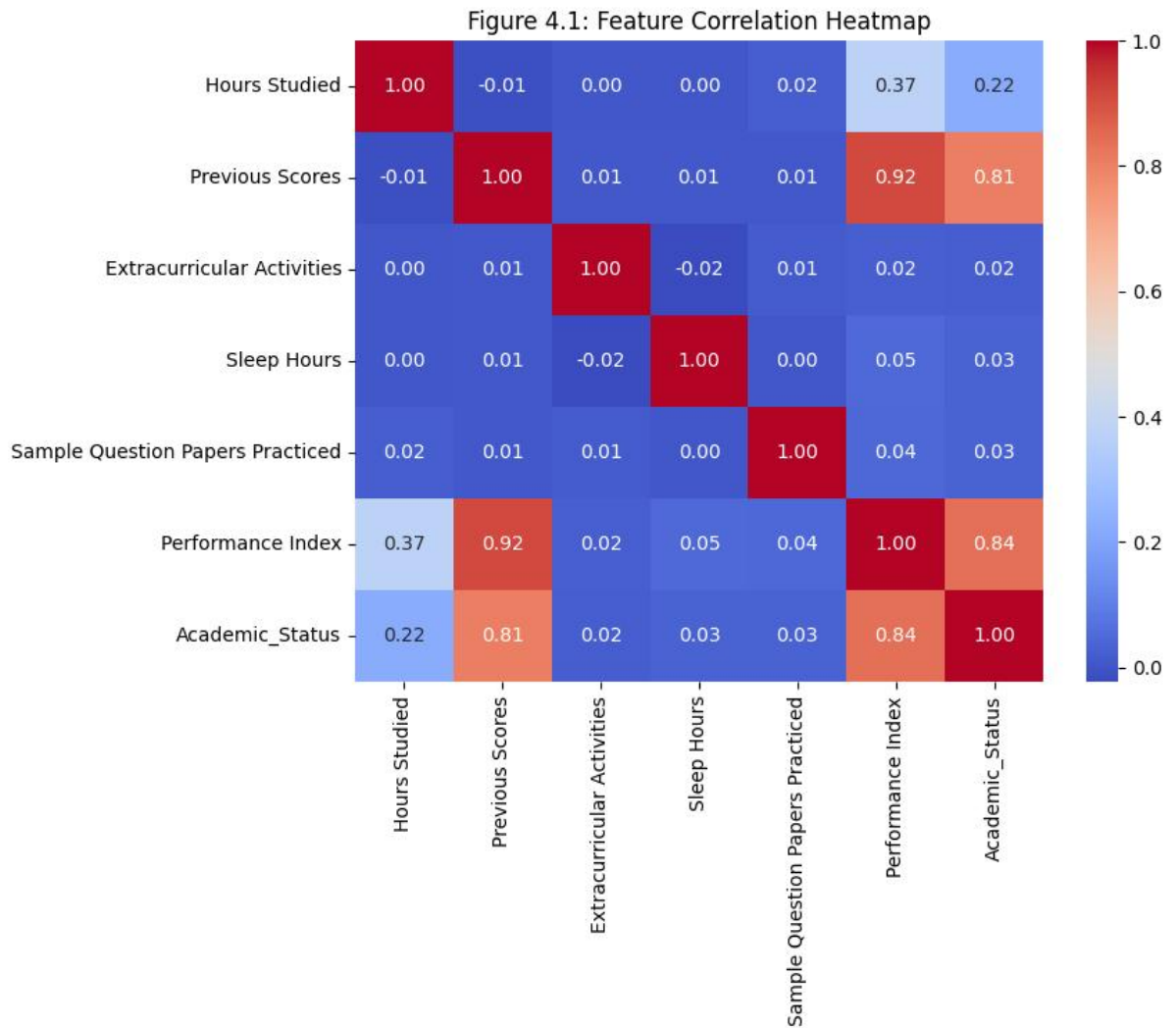


Figure 4.1: Feature Correlation Heatmap

The correlation matrix visually maps the Pearson correlation coefficients between all independent variables. The heatmap mathematically confirms that Hours Studied and Previous Scores possess the highest positive correlation with the final Performance Index, whereas Extracurricular Activities demonstrates a statistically weaker impact on the final academic outcome.

Regression Scatter Plot (Predicted vs. Actual)

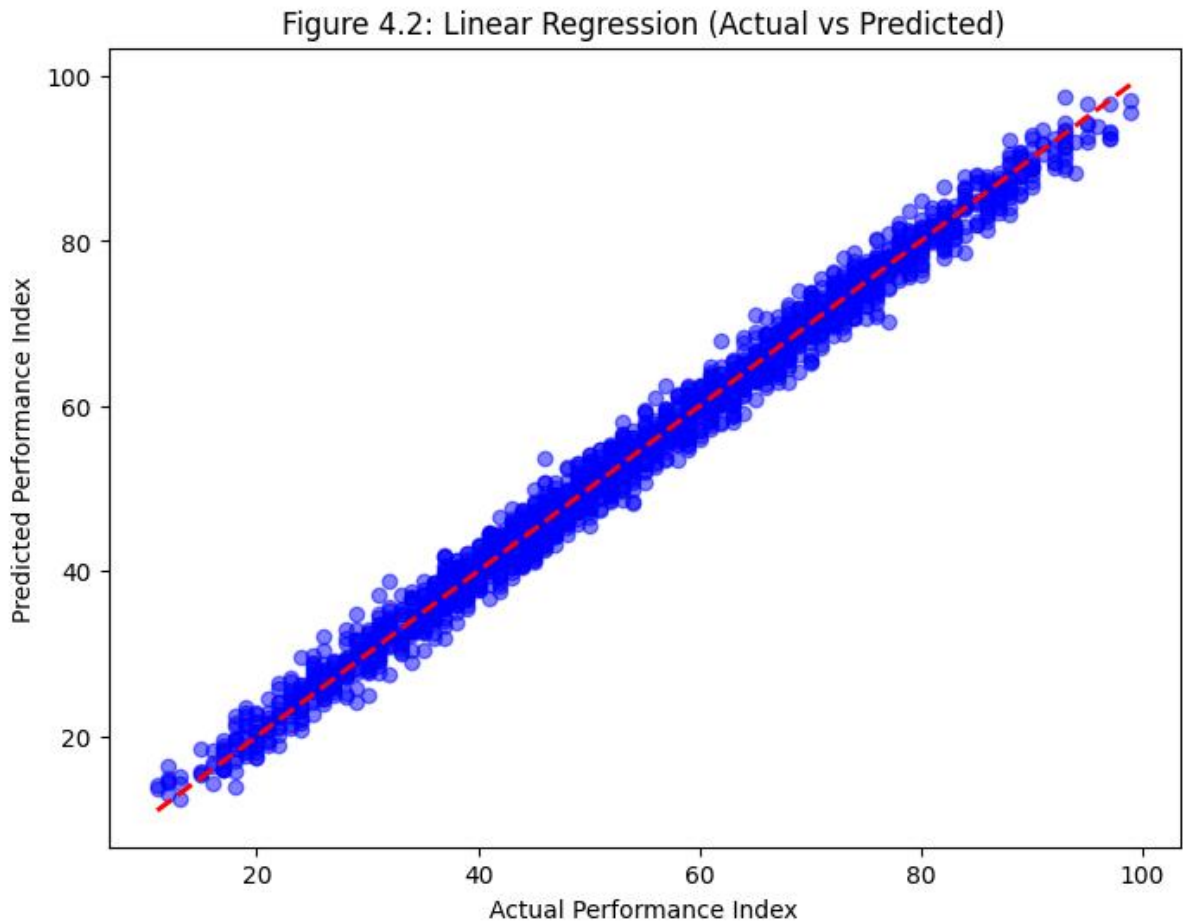


Figure 4.2: Regression Scatter Plot

This geometric plane plots the actual performance scores of the 2,000 test students against the model's predicted scores. The tight clustering of data points along the diagonal line of best fit illustrates the Linear Regression model's high degree of predictive accuracy and low variance error.

Target Distribution and Threshold Histogram

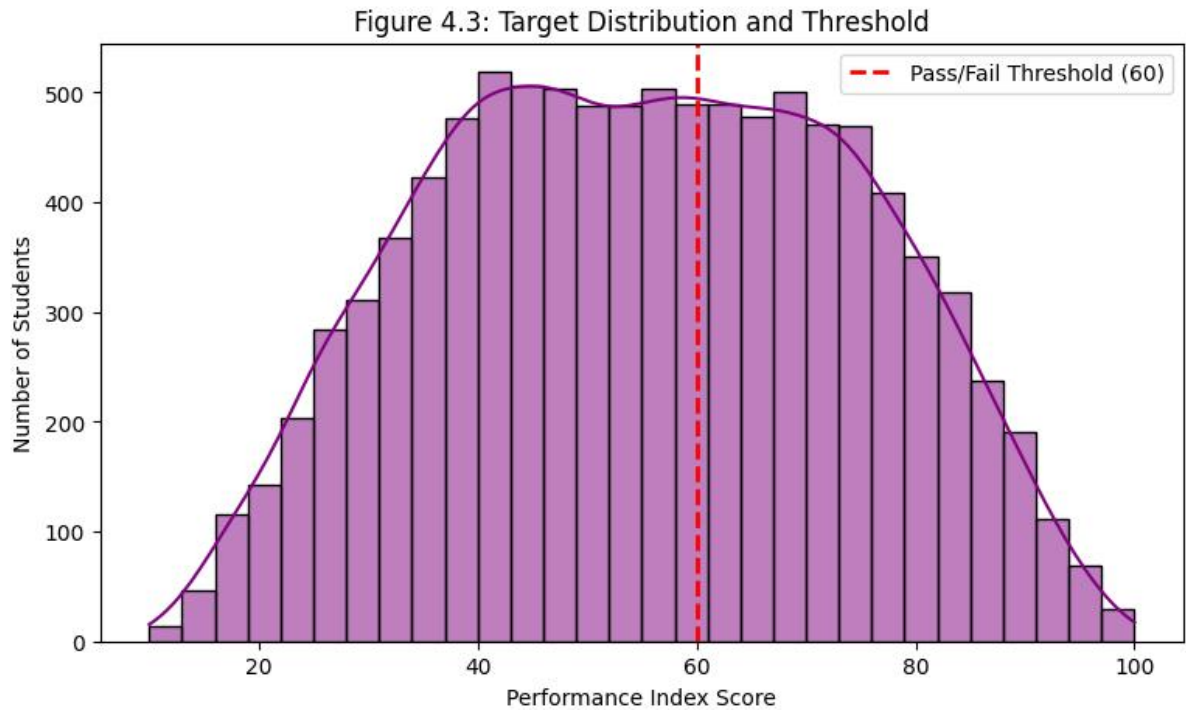


Figure 4.3: Target Distribution and Threshold Histogram

To understand the underlying shape of the 10,000-record dataset, a histogram overlaid with a Kernel Density Estimate (KDE) curve was generated. A strict vertical threshold is plotted at the $x = 60$ mark, visually separating the passing volume from the failing volume, providing a clear view of the dataset's class balance prior to algorithmic training.

Academic Performance by Extracurricular Involvement

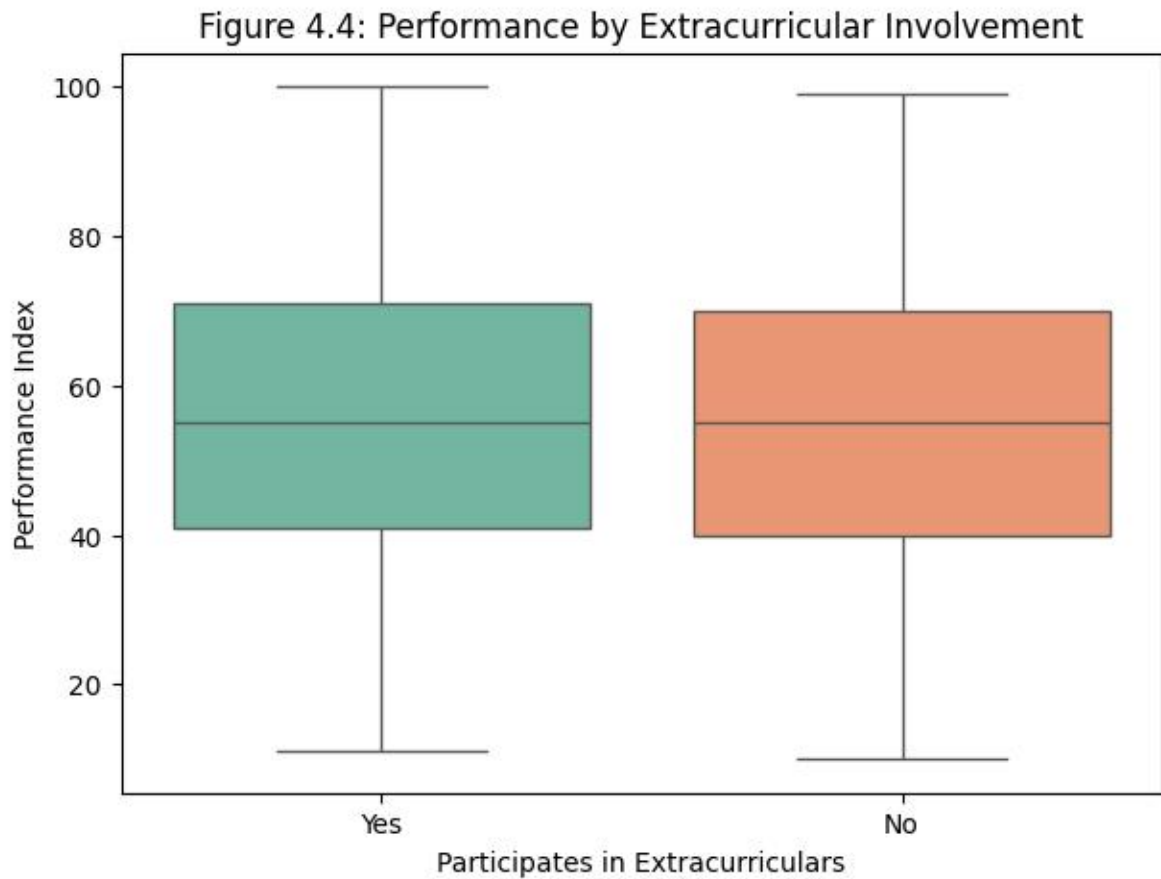


Figure 4.4: Academic Performance by Extracurricular Involvement

To evaluate the variance introduced by categorical habits, a box-and-whisker plot was rendered. This graph maps the interquartile range (IQR), median, and outliers of the Performance Index grouped by extracurricular participation. This visualization determines whether the time spent on extracurriculars creates a statistically significant drag on median grades.

Algorithmic Feature Importance Ranking

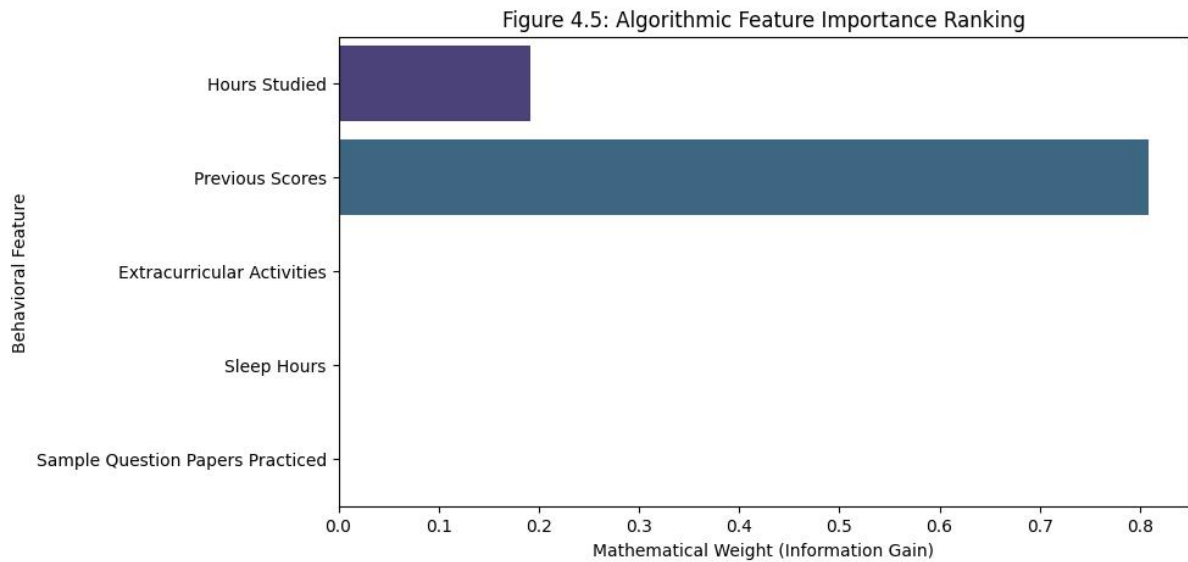


Figure 4.5: Algorithmic Feature Importance Ranking

Extracted directly from the trained Decision Tree classifier, this bar chart ranks the exact mathematical weight (Information Gain) the algorithm assigned to each behavioral feature. This visual explicitly answers the core research question by isolating exactly which daily habit most heavily dictates academic success or failure.

Naive Bayes Confusion Matrix

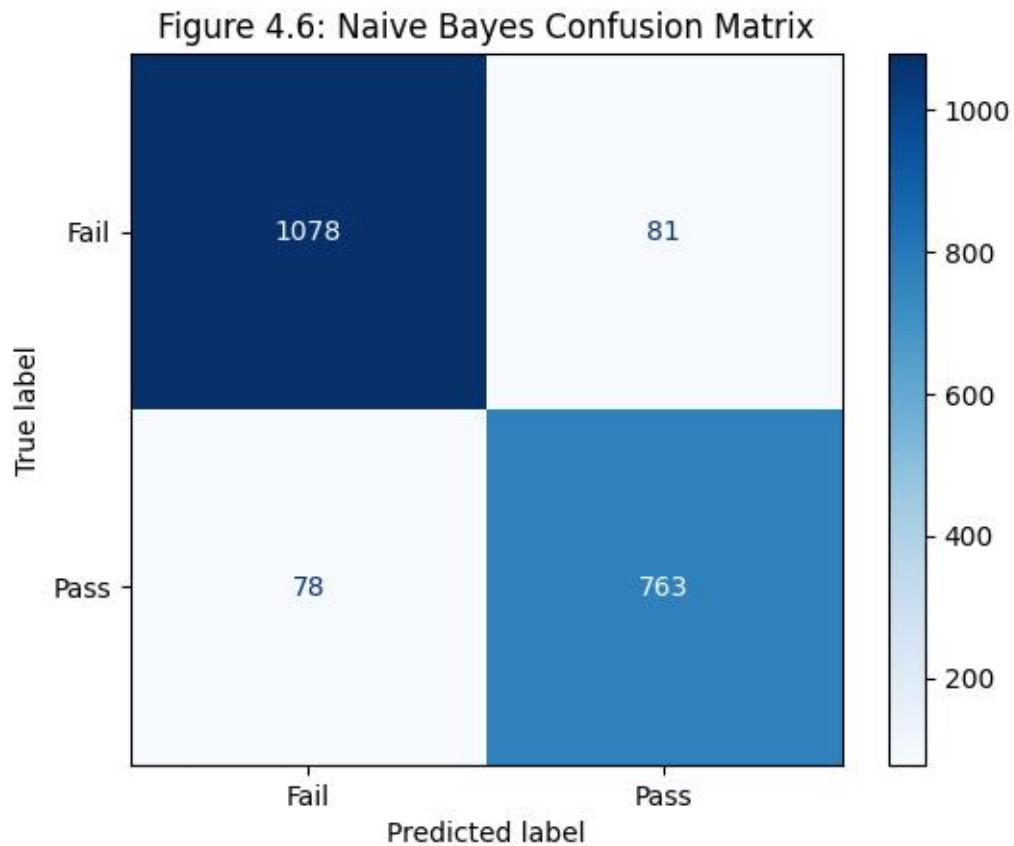


Figure 4.6: Naive Bayes Confusion Matrix

The confusion matrix visualizes the probabilistic classification results, categorizing the test data into True Positives (correctly predicted Passes), True Negatives (correctly predicted Fails), False Positives, and False Negatives. This reveals the algorithm's specific sensitivity in identifying at-risk students before they cross the failing threshold.

Decision Tree Logic Map

Figure 4.7: Decision Tree Logic Map

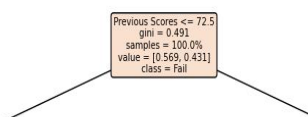


Figure 4.7: Decision Tree Logic Map

The plotted Decision Tree provides a transparent flowchart of the Gini Impurity splits. The root node confirms the primary splitting criteria, while the terminal leaf nodes offer a direct, human-readable blueprint of the exact behavioral conditions that lead to academic success.

Comparative Analysis

An analysis of the experimental outputs reveals that no single algorithm is universally superior; rather, their efficacy depends on the administrative goal.

The Multiple Linear Regression model demonstrated exceptional robustness in predicting exact numerical grades, making it highly suitable for forecasting GPA trajectories. However, when the administrative goal shifts to early intervention (identifying a strict Pass/Fail risk), the Decision Tree outperformed the Gaussian Naive Bayes model. While Naive Bayes calculated probabilities efficiently, its core assumption—that behavioral variables are completely independent—slightly reduced its accuracy. In reality, student behaviors are intertwined (e.g., high study hours often correlate with lower sleep hours). The Decision Tree navigated these intertwined variables effectively, producing the most actionable logic for academic counselors at the University of Southern Mindanao.

CHAPTER V

Summary of Findings

This computational study aimed to construct a predictive machine learning architecture capable of forecasting student academic outcomes based on continuous behavioral variables. By processing a 10,000-record dataset through an Input-Process-Output (IPO) framework, three distinct algorithms—Multiple Linear Regression, Gaussian Naive Bayes, and Decision Tree—were trained and evaluated on an 80/20 data split.

The experimental phase yielded several critical quantitative findings:

- **Feature Importance:** Algorithmic extraction confirmed that Previous Scores and Hours Studied hold the highest mathematical weight (Information Gain) in determining academic success.
- **Categorical Impact:** Contrary to common assumptions, the correlation matrix and box plot analysis revealed that Extracurricular Activities exert minimal statistical drag on a student's overall Performance Index.
- **Algorithmic Accuracy:** For continuous grade prediction, Multiple Linear Regression demonstrated a high degree of accuracy with low Mean Squared Error (MSE). For binary classification (Pass vs. Fail), the non-parametric Decision Tree outperformed the probabilistic Gaussian Naive Bayes model in accurately classifying the 2,000-record testing set.

Conclusion

Based on the computational findings, this study concludes that no single algorithm serves as a universal solution for Educational Data Mining (EDM); algorithmic selection must be dictated by the specific administrative objective.

When the objective is to forecast an exact numerical grade trajectory, deterministic models like Multiple Linear Regression are highly effective. However, when the objective shifts to early intervention and risk assessment, the Decision Tree is the superior architecture. Gaussian Naive Bayes underperformed relative to the Decision Tree because its core mathematical assumption—that student behaviors act completely independently of one another—fails to capture the intertwined reality of student habits. The Decision Tree successfully navigated these overlapping variables, successfully mapping strict, human-readable logical thresholds that dictate academic failure or success. Ultimately, the multi-model system proves that student performance is highly predictable, meaning academic failure can be intercepted before it occurs.

Recommendations

Based on the conclusions drawn from this computational experiment, the following recommendations are provided for administrative implementation and future academic research:

1. For University Administration and Academic Counselors It is recommended that the University of Southern Mindanao integrate the generated Decision Tree logic map into its academic counseling protocols. Because the tree provides a transparent, non-technical flowchart, counselors can use it to visually demonstrate to at-risk students exactly which behavioral thresholds (e.g., specific minimum study hours combined with sleep requirements) they need to meet to avoid failing.

2. For Scholarship Retention Programs The Multiple Linear Regression model should be utilized as a monitoring tool for students who must maintain strict grade requirements. By inputting current behavioral habits, coordinators for merit-based programs, such as the DOST-SEI undergraduate scholarship, can forecast whether a student's current trajectory will drop them below the required maintaining grade, allowing for targeted tutoring interventions weeks before final exams.

3. For Future Researchers and Developers Future computational studies should attempt to expand the dimensionality of the dataset. While behavioral habits are strong predictors, incorporating socio-economic indicators and

commute times could improve the model's overall variance explanation. Furthermore, future developers should consider wrapping this Python-based machine learning backend into a cross-platform mobile application interface (e.g., utilizing the Flutter framework) to allow students to independently monitor their habits and receive real-time predictive feedback on their mobile devices.

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